
Command Line Utilities

The service utilities available via the command line include the following.

How to Add/Change the VIP Address

Overview of VIP Addresses

A VIP is a single network connection defined on an HA (dual-server node) system to point to for server connections, regardless of which server node is currently operating as the primary (active) server node. While the VIP is configured on the primary node, using a NIC interface, when it is configured, it automatically saves a copy of the VIP configuration information to the other nodes in the StorNext cluster (typically the other server in the HA system). The VIP is configured on a specific interface, meaning that, even though the initial configuration settings are specific to the primary node they are configured on, when the file updates are made to the other clustered server(s), the updates to the VIP settings are assigned to the same interface on the other clustered server(s).

Example (HA server pair):

The VIP is updated to use the MAC address, IP and netmask for the eth0 port (interface) on the primary server (node 1). When the file is saved, the VIP settings saved on the primary are replicated to node 2, using the same interface (in this case, eth0) for node 2, but using its' unique MAC address for eth0, but with the same IP and netmask settings as were used for the primary server node.

Add or change the VIP address

To add or modify the Virtual IP address for the system, log in to the server node operating as primary:

1. Open an SSH connection to the appropriate server and use the IP address assigned to the node on the Management or LAN Client network, or use the Service Port IP address.

Note: The service port is no longer a dedicated service port. This port might be configured by the user for use as a metadata or client port with a different IP configuration. If this port has been reconfigured by the user, then command line access over a network connection requires an appropriate IP address for your configured system on that network.

Service Port IP addresses:

- Node 1 : *<service port IP address>* (dual-server node systems), the default service port IP address is 10.17.21.1
- Node 2 : *<service port IP address>* (dual-server node, or supported "HA-ready" single-server node StorNext appliances), the default service port IP address is 10.17.21.2

[close section](#)

2. Initiate an ssh session to the system using Terminal or PuTTY:

To ssh into the system

- a. Log in to the command line using the following credentials:

- User name: **stornext**
- Password: *<StorNext user account password>*

Note: **password** is the default password for the **stornext** user account. If the password has been changed, use the current password. The first time you log in, you are prompted to change the password to a different one.

- b. Enter **sudo rootsh** to gain root user access.
 - c. Enter the password for the **stornext** user account again.
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3. Verify that the server node you are connected to is operating as primary. Enter:

```
snhamgr -m status
```

4. Verify the output is (in green):

```
LocalMode=default
LocalStatus=primary
RemoteMode=default
RemoteStatus=running
```

This indicates the "Local" node you are connected to is operating as primary ("LocalStatus=primary"), and that the other node is operating as secondary ("RemoteStatus=running"). If you do not see this status, close this PuTTY/terminal session and SSH in to the other server node.

5. Obtain information for the interface you wish to assign for use as a VIP. Enter:

```
vip_control -i
```

If you see output similar to the following, there is a VIP assigned to the system.

Example (VIP exists):

```
NICs: name 'eth0' mac 00155db5e60b desc '' machdep '' IP 192.168.0.188
255.255.255.0
```

In this example, **eth0** has been configured as the VIP, using the default IP address and netmask assigned to this interface during initial configuration. If you are happy with this configuration, you can exit this SSH session. However, if you want to add another interface as another VIP for the system, continue to the next step.

Example (VIP does not yet exist):

If you see this message, there currently is no VIP assigned to the system (HA server pair):

```
Unable to read VIP file
```

Followed by information similar to the following:

```
NICs:
name 'p2p1' mac a0369f779b48 desc '' machdep ''
name 'p2p2' mac a0369f779b49 desc '' machdep ''
```

```

name 'p2p3' mac a0369f779b4a desc '' machdep ''
name 'p2p4' mac a0369f779b4b desc '' machdep ''
  IP 10.20.86.75 255.255.252.0
name 'bond0' mac ecf4bbdecabd6 desc '' machdep ''
  IP 192.168.1.11 255.255.255.0
name 'em1' mac ecf4bbdecabd4 desc '' machdep ''
  IP <service port IP address> 255.255.255.0
name 'em2' mac ecf4bbdecabd5 desc '' machdep ''
  IP 10.20.86.71 255.255.252.0
name 'idrac' mac 74e6e2fc2237 desc '' machdep ''

```

Some things to note about this example:

- All the available NIC ports on the system are shown, along with information about each one.
 - Ports **p2p1** through **p2p3** have not yet been configured, so there is no IP address or netmask assigned to them yet, meaning they cannot be used as a VIP.
 - Notice that **em2** (management network), and **p2p4** are configured. Typically, this is the port you would define as the system VIP, but it could be any active, configured NIC port on the system.
6. Copy the information you will need to configure the VIP for the interface you will use (MAC address, IP address, netmask, IPv6 address [if used])
 7. Navigate to the correct directory. Enter:

```
cd /usr/cvfs/bin
```

8. To configure the VIP, or add another interface to the existing VIP file, enter the **vip_control** command, following the specific variables described here:

```
vip_control -u <vip settings>
```

Required <vip settings> Syntax

```
<vip settings> = "<settings for VIP 1>;<settings for VIP 2>";"
```

The values for the <vip settings> consist of the following comma-separated values: MAC address, IPV4 VIP, netmask, IPV6 VIP (optional), and then prefix length. Where no value is set, use commas to separate from the other values. A semi-colon separates first and second VIPs and closes the string, followed by an ending straight quotation mark.

Note: Adding a second, VIP 2, is optional, not required.

Pro Tip!

To view all the VIP commands available, enter **vip_control** on the command line, and you will be presented the options and their syntax.

Example: server node 1 with 1 VIP (typical):

(Using values for **em2** from the example output above that didn't yet have a VIP assigned to the system.)

```
vip_control -u "ecf4bbdecbd5,10.20.86.71,255.255.255.0,,;"
```

Example: server node 1 with 2 VIPs defined:

(Using values for **em2** and **p2p4** from the example output above that didn't yet have a VIP assigned to the system.)

```
vip_control -u "ecf4bbdecbd5,10.20.86.71,255.255.252.0,,;  
a0369f779b4b,10.20.86.75,255.255.252.0,,;"
```

Note: In these examples, IPV6 VIP and prefix length are not used for either VIP address, but the commas and semi-colons used to separate those fields are still included in the string.

You will see a note similar to the following that identifies that the settings are being written to the VIP file (in this case, from the eth2 update example [the first VIP example, above], since there was no VIP file this would create a new vip file). Then, the VIP is configured, and the configuration updates would be replicated to other server nodes in the StorNext cluster (the "Telling others about IP..." portion):

Writing new VIP file

```
Running '/sbin/ifconfig em2:ha 10.20.86.71 netmask 255.255.255.0 up'
```

```
Telling others about IP (/usr/cvfs/lib/takeover_ip -d em2:ha -m  
ecf4bbdecbd5 -i 10.20.86.71)
```

9. Update the firewall rules for the node operating as the primary. Enter:

```
/opt/DXi/scripts/netcfg.sh --reset_snvip
```

10. Close the SSH connection for the operating as the primary.

The procedure is complete.

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[close section](#)

How to Reset the StorNext User Password

This section explains how to boot the server node so that the **stornext** user password can be reset.

To reset the **stornext** user password:

1. Connect the service laptop to the BMC port and launch the BMC console. Refer to [Access the BMC](#) for more information.
2. Run the following command to reboot the server node and then wait for the GRUB menu to appear on the console for the Linux boot:

```
systemctl -i reboot
```

3. Select option **e** to prompt for the GRUB password.

4. Enter the GRUB username: **root**
5. Enter the GRUB password **Qa@Gr64!**
6. Use the arrow keys to move down to the line that begins with **linux**.
7. Add **<space>init=/bin/bash** to the end of the boot line and press **Ctrl-x** to continue the boot process.

Note: The system then boots to a Linux prompt without a password prompt.

8. Enter **passwd stornext** at the Linux prompt
9. Enter the factory default password of **password** when prompted.
10. Perform a reset of the node by navigating to the **Power Control** menu on the BMC console and select **Set Power Reset**.
11. On system reboot, have the customer set the **stornext** password to a unique value and record it somewhere before leaving the customer site.

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How to Set the IPMI Password

You can change the default factory password for the IPMI remote login/BMC user by selecting the **Set IPMI Password** option from the **Service Menu** and entering a custom password.

Note: The default password from the factory is available on the pull out tab on the front of the server.

Do the following to set the IPMI password:

1. Log in to the CLI of the server node that you are modifying:

To log in to the CLI

- a. Open an SSH connection to the appropriate server and use the IP address assigned to the node on the Management or LAN Client network.
- b. Initiate an ssh session to the system using Terminal or PuTTY:

To ssh into the system

- a. Log in to the command line using the following credentials:
 - User name: **stornext**
 - Password: *<StorNext user account password>*

Note: **password** is the default password for the stornext user account. If the password has been changed, use the current password. The first time you log in, you are prompted to change the password to a different one.

- b. Enter **sudo rootsh** to gain root user access.
- c. Enter the password for the **stornext** user account again.

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[close section](#)

2. At the prompt, enter the following to open the **Service Menu** as the **admin** user:

```
sh /opt/DXi/scripts/service.sh -admin
```

3. Select **Hardware Configuration > Setup IPMI > Set IPMI Password**.

Example

*** Service Menu ***

```
0) Hardware Configuration  >>> - Tools for Configuring Hardware.
1) Serial Numbers         >>> - Tools for Node and System Serial Numbers.
2) More Tools             >>> - Miscellaneous Tools.
```

Your Choice ('q' to Exit this Menu): 0

*** Service Hardware Configuration Menu ***

```
0) Setup IPMI              >>> - Setup the iDRAC/IPMI.
1) Factory Detect Hardware - Create Factory List of Detected Hardware.
```

Your Choice ('q' to Exit this Menu): 0

*** Service Setup IPMI Menu ***

```
0) Configure IPMI      - Configure IPMI network Information.
1) Show IPMI           - Show IPMI network Information.
2) Set IPMI Password   - Set IPMI remote login password.
3) Reboot IPMI         - Reboot the IPMI controller.
```

Your Choice ('q' to Exit this Menu): 2

Enter the new password for the IPMI remote login.
The password must be at least 8 characters and contain a lowercase letter, an uppercase letter, and a non-alpha character.
To exit, press ENTER when prompted for the password.

Enter password:

4. Enter a password for the IPMI remote login/BMC user.
5. Exit the **Service Menu** session.
6. Exit your **rootsh** session.
7. Exit the SSH connection to the server.

[close section](#)

How to Reset the IMPI/BMC Password to the Factory Default

This section provides information on how to reset the IMPI/BMC password to the factory default.

Note: The default password from the factory is available on the pull out tab on the front of the server.

Do the following to reset the IMPI/BMC password to the factory default:

1. Log in to the CLI of the server node that you are modifying:

To log in to the CLI

- a. Open an SSH connection to the appropriate server and use the IP address assigned to the node on the Management or LAN Client network.
- b. Initiate an ssh session to the system using Terminal or PuTTY:

To ssh into the system

a. Log in to the command line using the following credentials:

- User name: **stornext**
- Password: *<StorNext user account password>*

Note: **password** is the default password for the stornext user account. If the password has been changed, use the current password. The first time you log in, you are prompted to change the password to a different one.

b. Enter **sudo rootsh** to gain root user access.

c. Enter the password for the **stornext** user account again.

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2. At the prompt, enter the following:

Note: After you execute the following command, the IPMI/BMC will reboot.

```
/opt/smc/bin/ipmicfg -fd 2
```

Example Output

Reset to the factory default completed.

[close section](#)

How to use the Wipe/Restore Utility

You can use the wipe/restore utility to wipe the StorNext install and HA shared file system off of StorNext Appliances. The utility then restores the appliance to factory state including the HA shared file system and StorNext without having to reinstall StorNext from DVD. This utility also preserves network configuration settings, so network configuration will be left intact and will be used by the restored system.

Note: This utility only applies to StorNext Appliances.

Caution: Only run this utility when you are not actively using the system.

On the Secondary Node

To use the wipe/restore utility, begin by accessing the server Node operating as the secondary:

1. Open an SSH connection to the appropriate server and use the IP address assigned to the node on the Management or LAN Client network, or use the Service Port IP address.

Service Port IP addresses:

- Node 1 : *<service port IP address>* (dual-server node systems), the default service port IP address is 10.17.21.1
- Node 2 : *<service port IP address>* (dual-server node, or supported "HA-ready" single-server node StorNext appliances), the default service port IP address is 10.17.21.2

[close section](#)

2. Initiate an SSH session to the system using Terminal or PuTTY:

a. Log in to the command line using the following credentials:

- User name: **stornext**
- Password: *<StorNext user account password>*

Note: **password** is the default password for the stornext user account. If the password has been changed, use the current password. The first time you log in, you are prompted to change the password to a different one.

- b. Enter **sudo rootsh** to gain root user access.
 - c. Enter the password for the **stornext** user account again.
3. At the command prompt type:

sh /opt/DXi/scripts/reset_SN.sh

[close section](#)

On the Primary Node

Continue by logging onto the MDC Node operating as the primary:

1. Open an SSH connection to the appropriate server and use the IP address assigned to the node on the Management or LAN Client network, or use the Service Port IP address.

Service Port IP addresses:

- Node 1 : *<service port IP address>* (dual-server node systems), the default service port IP address is 10.17.21.1
- Node 2 : *<service port IP address>* (dual-server node, or supported "HA-ready" single-server node StorNext appliances), the default service port IP address is 10.17.21.2

[close section](#)

2. Initiate an SSH session to the system using Terminal or PuTTY:
 - a. Log in to the command line using the following credentials:
 - User name: **stornext**
 - Password: *<StorNext user account password>*

Note: **password** is the default password for the stornext user account. If the password has been changed, use the current password. The first time you log in, you are prompted to change the password to a different one.

- b. Enter **sudo rootsh** to gain root user access.
 - c. Enter the password for the **stornext** user account again.
3. From the command prompt, enter:

sh /opt/DXi/scripts/reset_SN.sh

[close section](#)

Convert to HA and restart the system

1. Convert the system from a single-node mode to an HA configuration.
2. Restart the metadata array and MDC nodes.

[close section](#)

[close section](#)

Core Files and the Trimcores Cron Job Process

The trimcores cron job process runs hourly on both MDC nodes and checks for new core files located in the /scratch/core/ directory on the node. If a new core file is detected, a RAS ticket is generated to report the core file, and the core file name is changed to **<corefile>-reported**.

Core files remain on the system for 30 days before they are deleted from the system. After 30 days, an empty text file named **<corefile>-deleted** remains in the /scratch/core/ directory to provide evidence that a core file was detected and deleted by the trimcores cron job process.

[close section](#)

Troubleshooting Tools

The following tools may be used for troubleshooting the appliance in the field.

Tool Name	Location	Use
iozone	/opt/iozone/bin/iozone	This is a file system benchmark utility, used for measuring file I/O.
netperf	/usr/bin/netperf	Network bandwidth testing. This tool times the transmission and reception of data between two systems using the User Datagram Protocol (UDP) or Transmission Control Protocol (TCP) protocols.

[close section](#)

Send us your comments

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